

MeetingOps Platform - Database Architecture & Schema Specification

Conforming to: Software Requirements Specification (SRS) Version 2.0 (Final)

1. Executive Summary & Compliance Overview

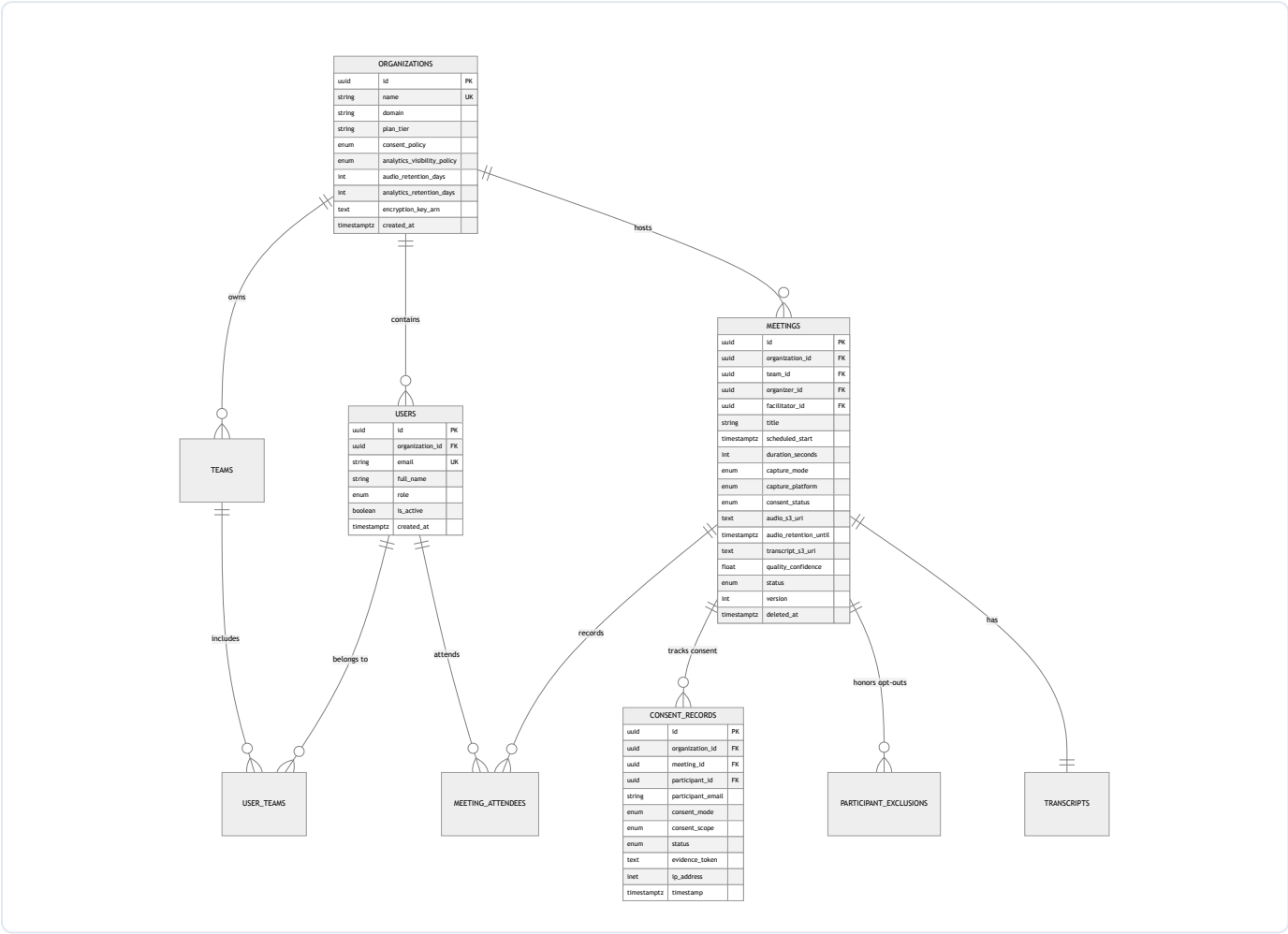
This specification defines the complete, production-grade relational and vector database architecture for **MeetingOps** (v2.0). Designed on **PostgreSQL 16+** with the **pgvector** and **pgcrypto** extensions, this architecture strictly enforces multi-tenant isolation, ethical AI constraints, Human-in-the-Loop (HITL) audit trails, consent verification, and privacy-preserving team analytics.

1.1 SRS v2.0 Requirements Traceability Matrix

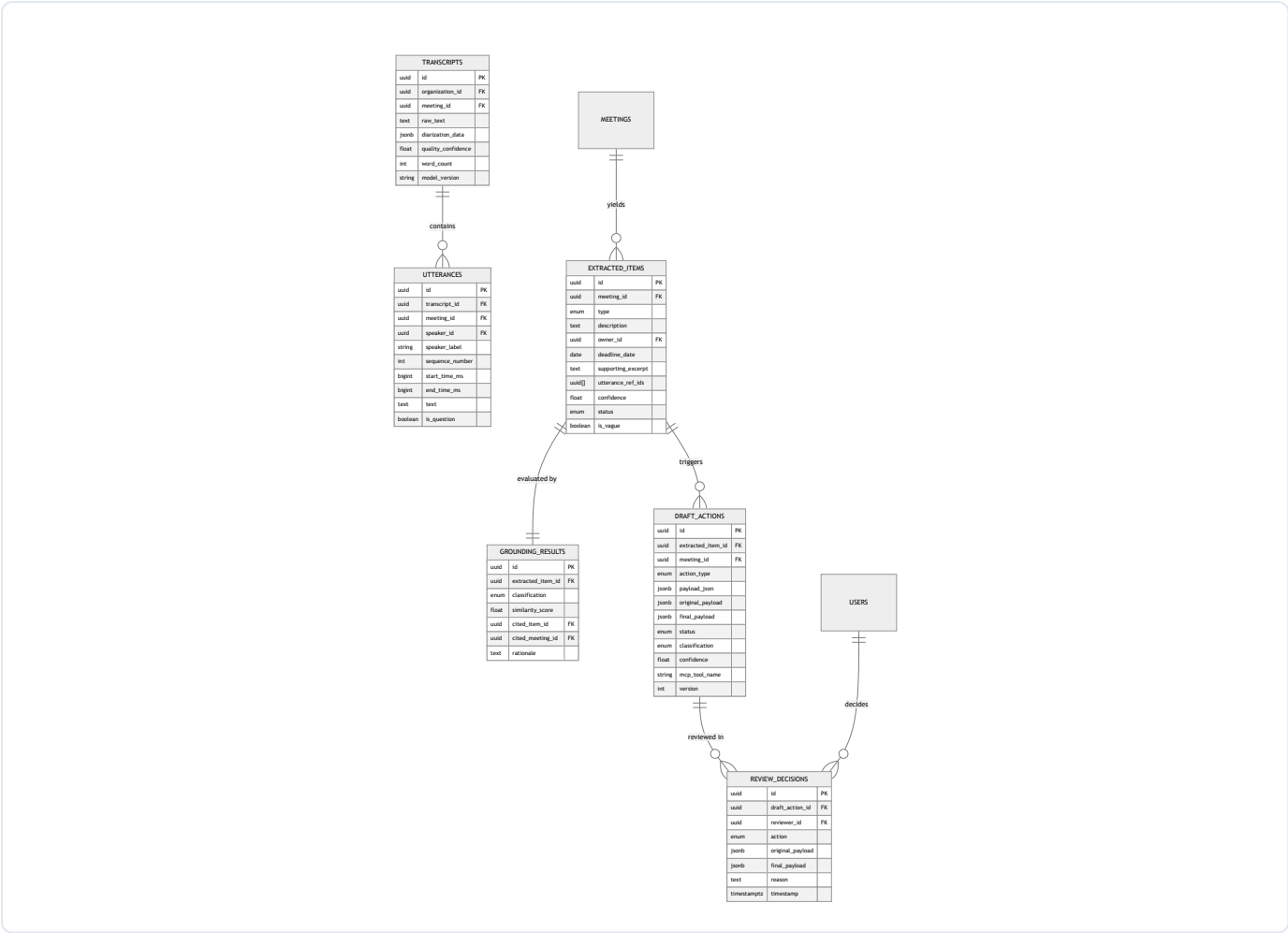
SRS Requirement	Subsystem / Policy	Database Implementation & Protection Mechanism
FR-1.1 – FR-1.6	Meeting Capture (Auto & Manual)	<code>meetings</code> , <code>meeting_attendees</code> , <code>participant_exclusions</code> , <code>consent_status</code> state machine
FR-2.1 – FR-2.5	Transcription & Diarization	<code>transcripts</code> , <code>utterances</code> (millisecond timestamp bounds), <code>voice_biometric_profiles</code>
FR-3.1 – FR-3.3	Structured Extraction	<code>extracted_items</code> with confidence scores, non-fabrication nullability, and prompt versioning
FR-4.1 – FR-4.2	Historical Grounding (RAG)	<code>grounding_results</code> , <code>meeting_embeddings</code> with pgvector HNSW cosine indexing & tenant isolation
FR-5.1	Validation	<code>extracted_items.is_vague</code> , <code>status = 'NEEDS_CLARIFICATION'</code> , duplicate prevention logic
FR-6.1 – FR-6.2	Draft Actions & HITL Review	<code>draft_actions</code> , <code>review_decisions</code> immutable audit log (<code>APPROVE</code> , <code>EDIT</code> , <code>REJECT</code>)
FR-7.1 – FR-7.7	Conversational Q&A Assistant	<code>qa_sessions</code> , <code>qa_queries</code> with cited transcript excerpts, latency, tokens, and audit logging
FR-8.1 – FR-8.8	Engagement Analytics	<code>meeting_participation_summaries</code> (team aggregate), <code>engagement_metrics</code> (individual), <code>v_secure_individual_engagement</code>
FR-8.3 / FR-9.1	Anti-Scoring & Anti-Ranking	Hard Architectural Prohibition: Zero score/rank columns; database check constraints; no leaderboards
FR-8.4 / FR-8.7	Privacy & Engagement Audit	<code>engagement_access_logs</code> (immutable), RLS & secure views restricting visibility to self/facilitator
FR-10.1 – 10.3	Multi-Tenancy & Consent Policy	PostgreSQL Row-Level Security (RLS) on all tenant tables; <code>organizations.consent_policy</code>
FR-10.4 – 10.5	Observability & Tracing	<code>agent_traces</code> capturing OpenTelemetry spans, token counts, and execution latencies
NFR-4.1 – 4.6	Security, Privacy & Biometrics	Encryption at rest, <code>voice_biometric_profiles</code> with explicit consent and retention TTL, ephemeral audio purge
PRIV-1 – PRIV-8	Consent & Responsible-Use	<code>consent_records</code> (immutable), <code>participant_exclusions</code> (FR-1.5 opt-out), synthetic demo tagging

2. Entity-Relationship Diagrams (ERD)

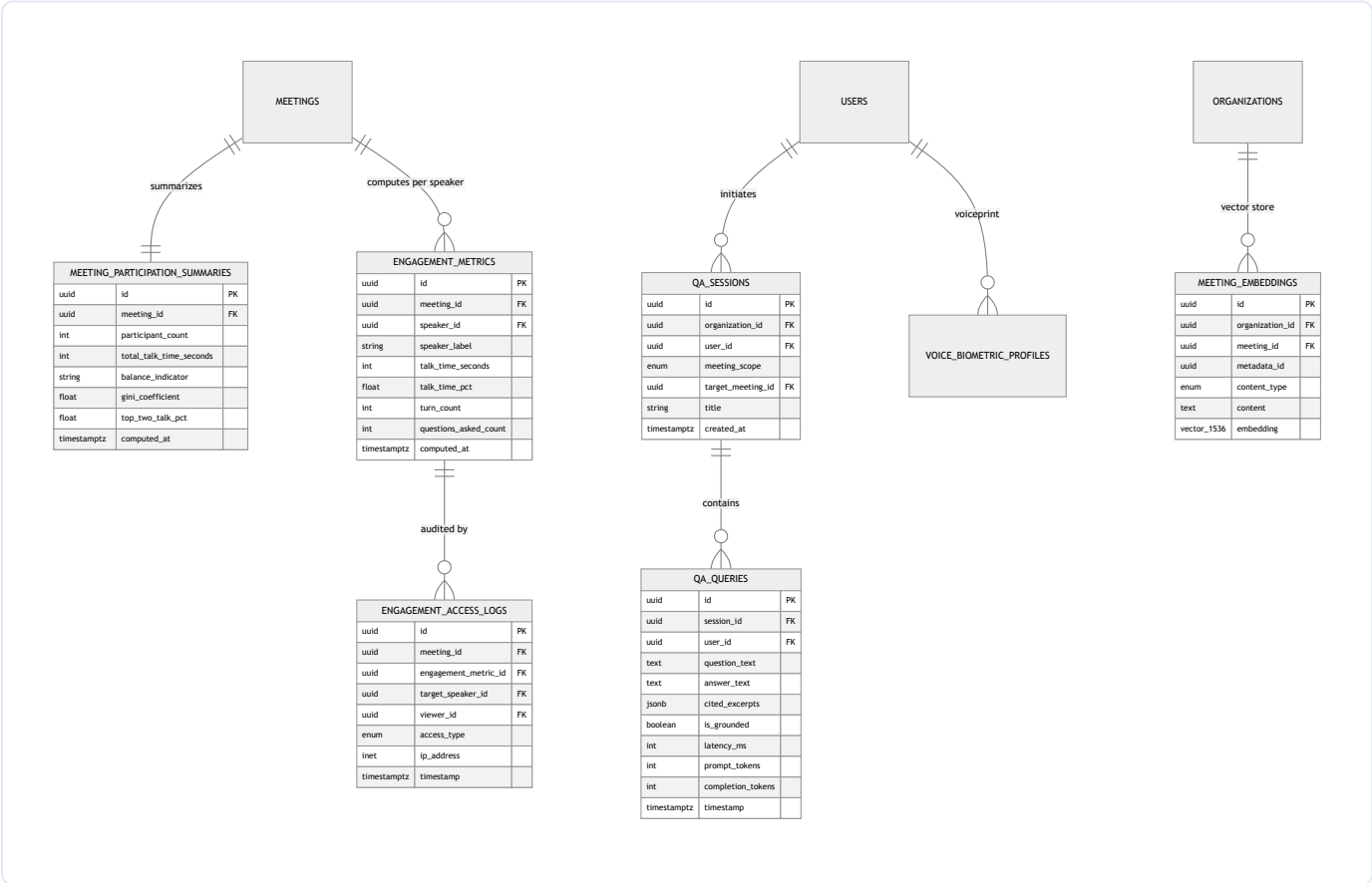
2.1 Core Multi-Tenancy, Identity, and Capture Subsystem



2.2 Diarization, Extraction, Grounding & HITL Review Pipeline



2.3 Conversational Q&A and Privacy-Preserving Engagement Analytics



3. Data Dictionary & Detailed Schema Specifications

3.1 organizations (Tenant Master)

- **Purpose:** Represents an enterprise tenant with organizational policies for recording consent and analytics visibility.
- **Columns:**
 - `id` (`UUID PRIMARY KEY`): Unique tenant identifier.
 - `name` (`VARCHAR(255) NOT NULL UNIQUE`): Organization legal/display name.
 - `domain` (`VARCHAR(255)`): Email domain for SSO routing.
 - `plan_tier` (`VARCHAR(50) NOT NULL DEFAULT 'community'`): Tier (`community` , `pro` , `enterprise`).
 - `consent_policy` (`consent_policy_enum NOT NULL DEFAULT 'notify_only'`): Configurable compliance mode (`notify_only` , `meeting_opt_in` , `participant_opt_in`) per **PRIV-1**.
 - `analytics_visibility_policy` (`analytics_visibility_policy_enum NOT NULL DEFAULT 'SELF_AND_FACILITATOR'`): Visibility boundary (`SELF_AND_FACILITATOR` , `SELF_ONLY` , `MANAGER_OVERRIDE_ENABLED`) per **FR-8.4**.
 - `audio_retention_days` (`INTEGER NOT NULL DEFAULT 0`): Raw audio TTL. `0` indicates immediate deletion post-transcription per **NFR-4.2**.
 - `analytics_retention_days` (`INTEGER NOT NULL DEFAULT 365`): Individual analytics retention period per **Section 6.2**.
 - `encryption_key_arn` (`TEXT`): Customer-managed KMS key reference for envelope encryption.
 - `is_active` (`BOOLEAN NOT NULL DEFAULT TRUE`): Tenant operational status.
 - `created_at` , `updated_at` (`TIMESTAMPTZ NOT NULL DEFAULT NOW()`).

3.2 users (Identity & RBAC)

- **Purpose:** System users mapped to organizational roles.
- **Columns:**
 - `id` (`UUID PRIMARY KEY`): User identifier.
 - `organization_id` (`UUID NOT NULL REFERENCES organizations(id) ON DELETE CASCADE`): Multi-tenant boundary.
 - `email` (`VARCHAR(255) NOT NULL`): Corporate email address (`UNIQUE(organization_id, email)`).
 - `full_name` (`VARCHAR(255) NOT NULL`): User display name.
 - `role` (`user_role_enum NOT NULL DEFAULT 'PARTICIPANT'`): `ORG_ADMIN` , `FACILITATOR` , `REVIEWER` , `PARTICIPANT` , `PLATFORM_ENGINEER` .
 - `avatar_url` (`TEXT`): User profile image URL.
 - `is_active` (`BOOLEAN NOT NULL DEFAULT TRUE`).
 - `created_at` , `updated_at` (`TIMESTAMPTZ NOT NULL DEFAULT NOW()`).

3.3 meetings (Meeting Lifecycle System of Record)

- **Purpose:** Tracks scheduled, recorded, uploaded, and processed meetings.
- **Columns:**
 - `id` (`UUID PRIMARY KEY`): Meeting identifier.
 - `organization_id` (`UUID NOT NULL REFERENCES organizations(id) ON DELETE CASCADE`): Tenant boundary.
 - `team_id` (`UUID REFERENCES teams(id) ON DELETE SET NULL`): Associated team.
 - `organizer_id` (`UUID REFERENCES users(id) ON DELETE SET NULL`): Organizer.
 - `facilitator_id` (`UUID REFERENCES users(id) ON DELETE SET NULL`): Designated meeting facilitator (granted analytics visibility per **FR-8.4**).
 - `title` (`VARCHAR(500) NOT NULL`): Meeting title.
 - `scheduled_start` (`TIMESTAMPTZ NOT NULL`): Scheduled meeting start.
 - `scheduled_end` , `actual_start` , `actual_end` (`TIMESTAMPTZ`).
 - `duration_seconds` (`INTEGER`): Total duration in seconds.
 - `capture_mode` (`meeting_capture_mode_enum NOT NULL DEFAULT 'MANUAL_UPLOAD'`): `AUTO_BOT` , `MANUAL_UPLOAD` , `DIRECT_INTEGRATION` .
 - `capture_platform` (`meeting_platform_enum NOT NULL DEFAULT 'CUSTOM_AUDIO'`): `ZOOM` , `GOOGLE_MEET` , `MS_TEAMS` , `SLACK_HUDDLE` , `CUSTOM_AUDIO` .

- `consent_status` (`meeting_status_enum` NOT NULL DEFAULT 'PENDING'): `PENDING`, `SATISFIED`, `BLOCKED_NO_CONSENT`, `NOT_APPLICABLE` (enforced before capture bot joins per **FR-1.3**, **PRIV-2**).
- `audio_s3_uri` (`TEXT`): Ephemeral object storage URI for raw audio.
- `audio_retention_until` (`TIMESTAMPZ`): Timestamp when audio must be permanently purged per **NFR-4.2**.
- `audio_deleted_at` (`TIMESTAMPZ`): Verification timestamp of audio deletion.
- `transcript_s3_uri` (`TEXT`): Transcripts archive reference.
- `quality_confidence` (`DOUBLE PRECISION`): STT overall confidence indicator (`0.0` - `1.0`) per **FR-2.4**.
- `status` (`meeting_status_enum` NOT NULL DEFAULT 'SCHEDULED'): Lifecycle state.
- `version` (`INTEGER` NOT NULL DEFAULT `1`): Optimistic concurrency control.
- `deleted_at` (`TIMESTAMPZ` NULL): Soft-delete marker.
- `created_at`, `updated_at` (`TIMESTAMPZ` NOT NULL DEFAULT `NOW()`).

3.4 `consent_records` (Immutable Recording Compliance Log)

- **Purpose:** Records explicit per-meeting or per-participant consent decisions to satisfy legal recording and analysis regimes per **FR-1.3**, **PRIV-1**, **PRIV-2**, **NFR-6.3**.
- **Immutability Guarantee:** Protected by database trigger `prevent_audit_log_tampering()` against `UPDATE` and `DELETE`.
- **Columns:**
 - `id` (`UUID PRIMARY KEY`).
 - `organization_id` (`UUID` NOT NULL REFERENCES `organizations(id)` ON DELETE CASCADE).
 - `meeting_id` (`UUID` NOT NULL REFERENCES `meetings(id)` ON DELETE CASCADE).
 - `participant_id` (`UUID` REFERENCES `users(id)` ON DELETE SET NULL).
 - `participant_email` (`VARCHAR(255)` NOT NULL).
 - `consent_mode` (`consent_policy_enum` NOT NULL): `notify_only`, `meeting_opt_in`, `participant_opt_in`.
 - `consent_scope` (`consent_scope_enum` NOT NULL): `RECORDING`, `TRANSCRIPTION`, `ANALYTICS`, `VOICE_MATCHING`.
 - `status` (`consent_record_status_enum` NOT NULL): `PENDING`, `GRANTED`, `DECLINED`, `REVOKED`.
 - `evidence_token` (`TEXT`): Cryptographic proof token or calendar response ID.
 - `ip_address` (`INET`): Network origin of the consent decision.
 - `user_agent` (`TEXT`): Client user agent.
 - `timestamp` (`TIMESTAMPZ` NOT NULL DEFAULT `NOW()`).

3.5 `participant_exclusions` (Analytics Opt-Out Registry)

- **Purpose:** Satisfies **FR-1.5** allowing any participant to request their speech segments be excluded from team analytics while preserving transcript integrity for extraction and grounding.
- **Columns:**
 - `id` (`UUID PRIMARY KEY`).
 - `organization_id` (`UUID` NOT NULL REFERENCES `organizations(id)` ON DELETE CASCADE).
 - `meeting_id` (`UUID` NOT NULL REFERENCES `meetings(id)` ON DELETE CASCADE).
 - `participant_id` (`UUID` REFERENCES `users(id)` ON DELETE SET NULL).
 - `participant_email` (`VARCHAR(255)` NOT NULL).
 - `exclude_analytics` (`BOOLEAN` NOT NULL DEFAULT `TRUE`).
 - `requested_at` (`TIMESTAMPZ` NOT NULL DEFAULT `NOW()`).
 - `UNIQUE(meeting_id, participant_email)`.

3.6 `transcripts` & `utterances` (Diarized Speech Segments)

- **`transcripts`:**
 - `id` (`UUID PRIMARY KEY`).
 - `organization_id` (`UUID` NOT NULL REFERENCES `organizations(id)` ON DELETE CASCADE).
 - `meeting_id` (`UUID` NOT NULL UNIQUE REFERENCES `meetings(id)` ON DELETE CASCADE).
 - `raw_text` (`TEXT` NOT NULL): Full transcript string.
 - `diarization_data` (`JSONB` NOT NULL DEFAULT `'{}'::jsonb`): Raw STT speaker segmentation data.

- `quality_confidence` (`DOUBLE PRECISION NOT NULL DEFAULT 1.0`): Quality score (FR-2.4).
- `language_code` (`VARCHAR(10) DEFAULT 'en'`).
- `word_count` (`INTEGER NOT NULL DEFAULT 0`).
- `model_version` (`VARCHAR(100)`): STT model ID (e.g. `whisper-large-v3`).
- **utterances** :
- `id` (`UUID PRIMARY KEY`).
- `organization_id` (`UUID NOT NULL REFERENCES organizations(id) ON DELETE CASCADE`).
- `transcript_id` (`UUID NOT NULL REFERENCES transcripts(id) ON DELETE CASCADE`).
- `meeting_id` (`UUID NOT NULL REFERENCES meetings(id) ON DELETE CASCADE`).
- `speaker_id` (`UUID REFERENCES users(id) ON DELETE SET NULL`): Mapped participant ID (FR-2.2).
- `speaker_label` (`VARCHAR(100) NOT NULL`): Speaker tag (e.g., `"Speaker 1"`, `"Priya S."`).
- `sequence_number` (`INTEGER NOT NULL`): Chronological order.
- `start_time_ms` (`BIGINT NOT NULL`), `end_time_ms` (`BIGINT NOT NULL`): Millisecond timestamps (FR-2.3).
- `text` (`TEXT NOT NULL`): Spoken sentence/paragraph.
- `confidence` (`DOUBLE PRECISION DEFAULT 1.0`).
- `is_question` (`BOOLEAN NOT NULL DEFAULT FALSE`): Heuristic question detection flag for analytics (FR-8.1).
- `CONSTRAINT chk_utterance_timing CHECK (end_time_ms ≥ start_time_ms)` .

3.7 `extracted_items` & `grounding_results`

- **extracted_items** :
- `id` (`UUID PRIMARY KEY`).
- `organization_id` (`UUID NOT NULL REFERENCES organizations(id) ON DELETE CASCADE`).
- `meeting_id` (`UUID NOT NULL REFERENCES meetings(id) ON DELETE CASCADE`).
- `type` (`extracted_item_type_enum NOT NULL`): `ACTION_ITEM` or `DECISION` (FR-3.1).
- `description` (`TEXT NOT NULL`): Item text.
- `owner_id` (`UUID REFERENCES users(id) ON DELETE SET NULL`): Identified owner (nullable per **FR-3.2**; no fabrication).
- `raw_owner_name` (`VARCHAR(255)`): Explicitly stated owner name in transcript.
- `deadline_date` (`DATE`), `raw_deadline_text` (`VARCHAR(255)`): Stated deadline (nullable per **FR-3.2**).
- `supporting_excerpt` (`TEXT NOT NULL`): Direct transcript quote evidence.
- `utterance_ref_ids` (`UUID[] DEFAULT '{}'`): Array of referenced utterance IDs.
- `confidence` (`DOUBLE PRECISION NOT NULL DEFAULT 0.0`): LLM extraction confidence (FR-3.3).
- `status` (`extracted_item_status_enum NOT NULL DEFAULT 'EXTRACTED'`): `EXTRACTED`, `NEEDS_CLARIFICATION`, `GROUNDED`, `VALIDATED`, `DRAFTED`, `DISCARDED` .
- `is_vague` (`BOOLEAN NOT NULL DEFAULT FALSE`): Validation flag (FR-5.1).
- `prompt_version`, `model_version` (`VARCHAR(100)`): Provenance tracking (Section 6.2).
- **grounding_results** :
- `id` (`UUID PRIMARY KEY`).
- `organization_id` (`UUID NOT NULL REFERENCES organizations(id) ON DELETE CASCADE`).
- `extracted_item_id` (`UUID NOT NULL UNIQUE REFERENCES extracted_items(id) ON DELETE CASCADE`).
- `classification` (`grounding_classification_enum NOT NULL`): `NEW`, `DUPLICATE`, `CONTINUATION`, `CONFLICT` (FR-4.1).
- `similarity_score` (`DOUBLE PRECISION`): Vector cosine similarity score against historical embedding.
- `cited_item_id` (`UUID REFERENCES extracted_items(id) ON DELETE SET NULL`): Prior item link.
- `cited_meeting_id` (`UUID REFERENCES meetings(id) ON DELETE SET NULL`): Prior meeting link.
- `rationale` (`TEXT NOT NULL`): Grounding reasoning output.
- `model_version` (`VARCHAR(100)`).

3.8 `draft_actions` & `review_decisions` (HITL Audit Trail)

- **draft_actions** :
- `id` (`UUID PRIMARY KEY`).
- `organization_id` (`UUID NOT NULL REFERENCES organizations(id) ON DELETE CASCADE`).
- `extracted_item_id` (`UUID NOT NULL REFERENCES extracted_items(id) ON DELETE CASCADE`).

- `meeting_id` (`UUID NOT NULL REFERENCES meetings(id) ON DELETE CASCADE`).
- `action_type` (`draft_action_type_enum NOT NULL`): `CREATE_TASK`, `CALENDAR_REMINDER`, `DRAFT_EMAIL` (FR-6.1).
- `payload_json` (`JSONB NOT NULL`): Draft MCP payload.
- `original_payload` (`JSONB NOT NULL`), `final_payload` (`JSONB`).
- `status` (`draft_action_status_enum NOT NULL DEFAULT 'DRAFTED'`): `DRAFTED`, `APPROVED`, `EDITED`, `REJECTED`, `EXECUTED`, `FAILED`.
- `classification` (`grounding_classification_enum NOT NULL DEFAULT 'NEW'`).
- `confidence` (`DOUBLE PRECISION NOT NULL DEFAULT 0.0`).
- `mcp_tool_name` (`VARCHAR(100) NOT NULL`): Target MCP tool name.
- `external_ref_id` (`VARCHAR(255)`): Resulting Jira/Linear/Calendar external ID upon execution.
- `executed_at` (`TIMESTAMPTZ`): Execution timestamp (only possible after human review per **FR-6.2**, **NFR-6.1**).
- `version` (`INTEGER NOT NULL DEFAULT 1`): Optimistic concurrency control.
- **review_decisions** (Immutable HITL Audit Log):
- `id` (`UUID PRIMARY KEY`).
- `organization_id` (`UUID NOT NULL REFERENCES organizations(id) ON DELETE CASCADE`).
- `draft_action_id` (`UUID NOT NULL REFERENCES draft_actions(id) ON DELETE CASCADE`).
- `reviewer_id` (`UUID NOT NULL REFERENCES users(id) ON DELETE RESTRICT`): Human reviewer actor.
- `action` (`review_decision_action_enum NOT NULL`): `APPROVE`, `EDIT`, `REJECT`, `BULK_APPROVE`.
- `original_payload` (`JSONB NOT NULL`), `final_payload` (`JSONB`).
- `reason` (`TEXT NOT NULL`): Reviewer rationale.
- `decision_metadata` (`JSONB DEFAULT '{}'::jsonb`).
- `timestamp` (`TIMESTAMPTZ NOT NULL DEFAULT NOW()`).
- **Protection:** Protected by `prevent_audit_log_tampering()` trigger against alteration or deletion.

3.9 `qa_sessions` & `qa_queries` (Conversational Meeting Assistant)

- **`qa_sessions`**:
- `id` (`UUID PRIMARY KEY`).
- `organization_id` (`UUID NOT NULL REFERENCES organizations(id) ON DELETE CASCADE`).
- `user_id` (`UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE`).
- `meeting_scope` (`qa_scope_enum NOT NULL DEFAULT 'SINGLE_MEETING'`): `SINGLE_MEETING`, `TEAM_MEETINGS`, `ORG_HISTORY` (FR-7.5).
- `target_meeting_id` (`UUID REFERENCES meetings(id) ON DELETE CASCADE`), `target_team_id` (`UUID REFERENCES teams(id) ON DELETE CASCADE`).
- `title` (`VARCHAR(255) NOT NULL DEFAULT 'New Conversation'`).
- `is_active` (`BOOLEAN NOT NULL DEFAULT TRUE`).
- `created_at`, `updated_at` (`TIMESTAMPTZ NOT NULL DEFAULT NOW()`).
- **`qa_queries`** (Audit-Logged Q&A Invocations):
- `id` (`UUID PRIMARY KEY`).
- `organization_id` (`UUID NOT NULL REFERENCES organizations(id) ON DELETE CASCADE`).
- `session_id` (`UUID NOT NULL REFERENCES qa_sessions(id) ON DELETE CASCADE`).
- `user_id` (`UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE`).
- `question_text` (`TEXT NOT NULL`): User input query.
- `answer_text` (`TEXT NOT NULL`): Generated response.
- `cited_excerpts` (`JSONB NOT NULL DEFAULT '[]'::jsonb`): Array of cited transcript moments and item IDs per **FR-7.3**.
- `is_grounded` (`BOOLEAN NOT NULL DEFAULT TRUE`): `FALSE` when "no relevant content found" fallback is triggered per **FR-7.4**.
- `latency_ms` (`INTEGER`), `prompt_tokens` (`INTEGER`), `completion_tokens` (`INTEGER`).
- `prompt_version`, `model_version` (`VARCHAR(100)`).
- `timestamp` (`TIMESTAMPTZ NOT NULL DEFAULT NOW()`).

3.10 meeting_participation_summaries & engagement_metrics (Privacy-Preserving Analytics)

- **meeting_participation_summaries** (Team-Level Descriptive Aggregates per **FR-8.2, FR-8.5**):
 - **id** (**UUID PRIMARY KEY**).
 - **organization_id** (**UUID NOT NULL REFERENCES organizations(id) ON DELETE CASCADE**).
 - **meeting_id** (**UUID NOT NULL UNIQUE REFERENCES meetings(id) ON DELETE CASCADE**).
 - **participant_count** (**INTEGER NOT NULL**): Number of attendees.
 - **total_talk_time_seconds** (**INTEGER NOT NULL**): Aggregate speaking duration.
 - **balance_indicator** (**VARCHAR(255) NOT NULL**): Factual descriptive framing (e.g., "2 of 6 participants accounted for 70% of talk time").
 - **gini_coefficient** (**DOUBLE PRECISION**): Statistical equality metric (**0.0** to **1.0**).
 - **top_two_talk_pct** (**DOUBLE PRECISION**): Percentage of time taken by the two longest speakers.
 - **descriptive_notes** (**TEXT**): Contextual framing notes per **FR-8.6**.
 - **computed_at** (**TIMESTAMPTZ NOT NULL DEFAULT NOW()**).
- **engagement_metrics** (Individual-Level Factual Metrics per **FR-8.1, FR-8.3, FR-8.4**):
 - **id** (**UUID PRIMARY KEY**).
 - **organization_id** (**UUID NOT NULL REFERENCES organizations(id) ON DELETE CASCADE**).
 - **meeting_id** (**UUID NOT NULL REFERENCES meetings(id) ON DELETE CASCADE**).
 - **speaker_id** (**UUID REFERENCES users(id) ON DELETE CASCADE**).
 - **speaker_label** (**VARCHAR(100) NOT NULL**): Diarized speaker name.
 - **talk_time_seconds** (**INTEGER NOT NULL DEFAULT 0**): Factual speaking seconds.
 - **talk_time_pct** (**DOUBLE PRECISION NOT NULL DEFAULT 0.0**): Percentage of meeting duration.
 - **turn_count** (**INTEGER NOT NULL DEFAULT 0**): Number of speaking turns.
 - **questions_asked_count** (**INTEGER NOT NULL DEFAULT 0**): Heuristic question count.
 - **computed_at** (**TIMESTAMPTZ NOT NULL DEFAULT NOW()**).
- **HARD ETHICAL CONSTRAINT (FR-8.3, FR-9.1, PRIV-4)**: Zero columns exist for performance scores, rankings, percentiles, or individual evaluation grades. Check constraints strictly enforce valid positive counts and percentages.
- **engagement_access_logs** (Access Audit Trail per **FR-8.7, NFR-6.2**):
 - **id** (**UUID PRIMARY KEY**).
 - **organization_id** (**UUID NOT NULL REFERENCES organizations(id) ON DELETE CASCADE**).
 - **meeting_id** (**UUID NOT NULL REFERENCES meetings(id) ON DELETE CASCADE**).
 - **engagement_metric_id** (**UUID REFERENCES engagement_metrics(id) ON DELETE CASCADE**).
 - **target_speaker_id** (**UUID REFERENCES users(id) ON DELETE SET NULL**): The user whose individual data was inspected.
 - **viewer_id** (**UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE**): The user who initiated the view.
 - **access_type** (**engagement_access_type_enum NOT NULL**): **VIEW_OWN** , **VIEW_FACILITATOR** , **VIEW_ADMIN_AUDIT** .
 - **ip_address** (**INET**), **user_agent** (**TEXT**).
 - **timestamp** (**TIMESTAMPTZ NOT NULL DEFAULT NOW()**).
- **Protection**: Protected by **prevent_audit_log_tampering()** trigger against modification or deletion.

3.11 meeting_embeddings & voice_biometric_profiles

- **meeting_embeddings** (pgvector HNSW Vector Store per **FR-4.1, FR-4.2**):
 - **id** (**UUID PRIMARY KEY**).
 - **organization_id** (**UUID NOT NULL REFERENCES organizations(id) ON DELETE CASCADE**): Tenant boundary.
 - **meeting_id** (**UUID NOT NULL REFERENCES meetings(id) ON DELETE CASCADE**).
 - **metadata_id** (**UUID NOT NULL**): Referenced entity ID (utterance, extracted item, or meeting).
 - **content_type** (**embedding_content_type_enum NOT NULL**): **UTTERANCE** , **ACTION_ITEM** , **DECISION** , **TRANSCRIPT_CHUNK** .
 - **chunk_index** (**INTEGER NOT NULL DEFAULT 0**).
 - **content** (**TEXT NOT NULL**): Text representation.
 - **embedding** (**vector(1536)**): Vector embedding vector.

- **Index:** HNSW Index `idx_embeddings_hnsw_cosine USING hnsw (embedding vector_cosine_ops) WITH (m = 16, ef_construction = 64)`.
- **voice_biometric_profiles** (Biometric-Adjacent Data Protection per **NFR-4.6, PRIV-5**):
- `id` (`UUID PRIMARY KEY`).
- `organization_id` (`UUID NOT NULL REFERENCES organizations(id) ON DELETE CASCADE`).
- `user_id` (`UUID NOT NULL UNIQUE REFERENCES users(id) ON DELETE CASCADE`).
- `voiceprint_hash` (`TEXT NOT NULL`): One-way cryptographic hash of speaker acoustic features.
- `voice_embedding` (`vector(512)`): Speaker voiceprint embedding.
- `consent_verified` (`BOOLEAN NOT NULL DEFAULT FALSE`): Mandatory consent check before matching.
- `retention_until` (`TIMESTAMPTZ NOT NULL`): Strict expiration deadline.
- `created_at`, `updated_at` (`TIMESTAMPTZ NOT NULL DEFAULT NOW()`).

3.12 agent_traces (Observability Spans)

- **Purpose:** OpenTelemetry distributed traces and LLM invocation spans across all pipeline agents per **FR-10.4, FR-10.5**.
- **Columns:**
- `id` (`UUID PRIMARY KEY`).
- `organization_id` (`UUID NOT NULL REFERENCES organizations(id) ON DELETE CASCADE`).
- `meeting_id` (`UUID NOT NULL REFERENCES meetings(id) ON DELETE CASCADE`).
- `trace_id` (`VARCHAR(64)`), `span_id` (`VARCHAR(32)`), `parent_span_id` (`VARCHAR(32)`).
- `agent_role` (`VARCHAR(50) NOT NULL`): `CAPTURE`, `TRANSCRIBER`, `EXTRACTION`, `GROUNDING`, `VALIDATION`, `DRAFTING`, `QA`, `ANALYTICS`.
- `step_index` (`INTEGER NOT NULL`).
- `tool_called` (`TEXT`), `tool_input` (`JSONB`), `tool_output` (`JSONB`).
- `reasoning_excerpt` (`TEXT`): Extracted Chain-of-Thought reasoning.
- `latency_ms` (`INTEGER`), `prompt_tokens` (`INTEGER`), `completion_tokens` (`INTEGER`).
- `timestamp` (`TIMESTAMPTZ NOT NULL DEFAULT NOW()`).
- **Protection:** Protected by `prevent_audit_log_tampering()` trigger against modification or deletion.

4. Security, Privacy & Compliance Architecture

4.1 Multi-Tenant Logical Isolation (RLS)

PostgreSQL Row-Level Security is active on every organization table. When an application service establishes a database transaction, it sets the current tenant context:

```
-- Executed per-request / per-connection:
SET LOCAL app.current_org_id = 'c38a4d78-b179-4d64-9844-3d027cf884c9';
SET LOCAL app.current_user_id = 'e76b2f12-0941-47a3-8321-70bf812c75a2';
SET LOCAL app.current_user_role = 'FACILITATOR';
```

All queries automatically apply the tenant isolation policy:

```
CREATE POLICY teams_tenant_isolation_policy ON teams
FOR ALL
USING (
    organization_id = NULLIF(current_setting('app.current_org_id', true), '')::uuid
    OR current_setting('app.bypass_rls', true) = 'true'
);
```

4.2 Data-Layer Privacy Protection for Individual Metrics (FR-8.4)

To ensure application bugs cannot leak peer engagement data, access to individual engagement metrics is restricted through the secure view `v_secure_individual_engagement`:

```
CREATE OR REPLACE VIEW v_secure_individual_engagement AS
SELECT
    em.id, em.organization_id, em.meeting_id, em.speaker_id,
    em.speaker_label, em.talk_time_seconds, em.talk_time_pct,
    em.turn_count, em.questions_asked_count, em.computed_at
FROM engagement_metrics em
JOIN meetings m ON em.meeting_id = m.id
WHERE (
    -- 1. Individual can view their own data
    em.speaker_id = NULLIF(current_setting('app.current_user_id', true), '')::uuid
    OR
    -- 2. Meeting Facilitator can view metrics for meetings they lead
    is_meeting_facilitator(em.meeting_id, NULLIF(current_setting('app.current_user_id', true), '')::uuid) = 1
    OR
    -- 3. Org Admin with explicit documented audit session
    (current_setting('app.current_user_role', true) = 'ORG_ADMIN' AND current_setting('app.audit_mode_enabled', true) = 'true')
);
```

4.3 Immutability of Audit Trails

All compliance records (`review_decisions` , `consent_records` , `engagement_access_logs` , `agent_traces`) enforce write-once immutability via database triggers:

```
CREATE OR REPLACE FUNCTION prevent_audit_log_tampering()
RETURNS TRIGGER AS $$
BEGIN
    RAISE EXCEPTION 'SECURITY VIOLATION: Records in table % are immutable and cannot be updated or deleted.';
END;
$$ LANGUAGE plpgsql;
```

4.4 Automated Ephemeral Audio & Retention Management

Scheduled background routines execute automated data purges to satisfy the minimal storage footprint requirement (NFR-4.2, PRIV-5):

```
-- Purges raw audio pointers once retention expires (Default: immediately after transcription)
SELECT purge_expired_raw_audio();

-- Purges voice biometrics past their retention date
SELECT purge_expired_voice_biometrics();
```

5. Vector Search (pgvector HNSW) & Grounding Performance

5.1 Hierarchical Navigable Small World (HNSW) Configuration

Vector embeddings use cosine distance indexing optimized for low-latency similarity searches within a tenant's vector namespace:

```
CREATE INDEX idx_embeddings_hnsw_cosine
ON meeting_embeddings
USING hnsw (embedding vector_cosine_ops)
WITH (m = 16, ef_construction = 64);
```

5.2 Scoped Grounding & Q&A Query Pattern

Grounding and Q&A vector retrievals always filter by `organization_id` and `content_type` prior to nearest-neighbor distance calculations:

```
-- RAG query strictly restricted to tenant boundary (FR-4.2)
SELECT
    me.metadata_id,
    me.content_type,
    me.content,
    1 - (me.embedding <=> :query_embedding) AS similarity_score
FROM meeting_embeddings me
WHERE me.organization_id = :org_id
    AND me.content_type IN ('DECISION', 'ACTION_ITEM')
ORDER BY me.embedding <=> :query_embedding
LIMIT 5;
```

6. Migration Execution Guide

1. The complete schema is packaged in Flyway migration script:

```
meeting-ops-backend/meeting-service/src/main/resources/db/migration/V3__srs_v2_complete_schema.sql
```

2. When starting the backend with Docker Compose:

```
bash cd meeting-ops-backend docker-compose up --build
```

3. Flyway will automatically apply `V1`, `V2`, and `V3` in order, creating all tables, indexes, triggers, and Row Level Security policies.